

CHAMPION AGDM, AB, Canada

WMO: 712170

Lat: 50.2953N

Lon: 113.3467W

Elev: 3264

StdP: 13.04

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.5	-13.8	-25.4	1.5	-19.1	-19.2	2.2	-13.5	29.6	36.1	26.3	33.6	5.6	90	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.4	87.4	60.5	83.6	59.8	80.2	58.8	64.8	78.4	62.8	77.1	61.0	75.9	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	87.4	70.5	57.6	79.9	67.7	55.5	74.0	65.0	31.6	78.6	30.0	76.8	28.6	76.1	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	23.8	20.7	DB	-26.8	93.4	6.3	4.3	-31.4	96.5	-35.1	99.0	-38.6	101.4	-43.2	104.5	(4)
(5)			WB	-27.1	68.6	6.3	2.4	-31.6	70.3	-35.3	71.7	-38.8	73.1	-43.3	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	20.1	18.6	29.9	40.1	50.4	57.4	63.4	61.9	53.2	41.4	28.2	18.5		
	DBStd	19.52	15.24	14.49	13.32	8.81	7.96	5.52	5.22	5.93	8.30	9.64	13.70	14.56		
	HDD50	4827	928	878	625	310	94	6	0	2	60	292	656	976		
	HDD65	9075	1393	1298	1089	747	455	236	92	131	359	730	1104	1441		
	CDD50	1320	0	0	1	13	107	227	415	371	157	26	3	0		
	CDD65	92	0	0	0	0	2	8	42	34	6	0	0	0		
	CDH74	2587	0	0	1	12	140	224	960	925	311	14	0	0		
	CDH80	787	0	0	0	1	25	46	310	314	90	1	0	0		
	WSAvg	9.2	9.4	8.2	9.8	10.2	10.2	9.7	8.1	8.1	8.8	9.6	9.7	8.8		
	PrecAvg	16.1	0.4	0.5	0.8	1.1	2.4	3.9	2.0	1.6	1.5	0.7	0.6	0.5		
(15) Precipitation	PrecMax	25.9	0.9	1.4	1.9	2.0	4.3	11.4	4.7	3.7	5.4	1.7	1.7	1.1		
	PrecMin	7.0	0.0	0.0	0.2	0.4	0.3	1.7	0.3	0.1	0.2	0.2	0.0	0.0		
	PrecStd	4.2	0.2	0.4	0.5	0.5	1.3	2.3	1.4	0.9	1.3	0.4	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	54.1	63.2	74.7	83.3	86.1	91.8	91.5	87.7	75.4	64.3	51.7		
		MCWB	41.8	41.0	45.5	50.7	56.2	61.9	61.7	60.3	57.9	52.2	47.8	41.1		
	2%	DB	46.7	47.0	57.7	67.9	78.2	79.9	87.5	87.3	82.5	69.3	55.7	46.3		
		MCWB	38.4	37.5	43.1	48.1	54.0	58.4	62.5	59.7	56.6	50.7	43.4	37.8		
	5%	DB	43.1	42.5	53.3	62.6	72.9	75.8	83.2	83.4	77.1	63.6	50.9	42.2		
		MCWB	36.1	35.1	40.8	45.3	52.2	57.8	61.7	59.4	55.2	48.1	41.1	35.4		
	10%	DB	39.9	38.3	48.5	57.2	68.2	72.1	79.4	79.4	71.6	58.3	46.1	38.6		
		MCWB	34.4	32.7	38.9	42.5	50.1	55.5	61.2	58.5	53.2	45.4	37.8	33.0		
	0.4%	WB	42.8	42.3	47.3	51.7	57.5	65.2	69.6	64.8	60.0	54.5	48.5	42.2		
		MCDB	50.4	52.4	60.5	72.7	77.1	77.3	80.5	80.6	79.1	71.7	61.7	49.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.8	38.1	44.1	48.8	55.3	61.9	66.3	62.5	57.6	51.7	44.6	38.4		
		MCDB	46.3	46.5	55.9	65.1	74.4	75.6	79.9	80.0	77.9	66.7	54.6	45.2		
	5%	WB	36.7	35.3	41.4	46.1	53.2	59.2	64.2	61.0	55.7	48.7	41.3	35.7		
		MCDB	42.4	41.7	52.0	61.0	70.2	71.5	77.6	78.1	74.9	62.0	50.4	41.7		
	10%	WB	34.5	32.7	38.8	43.5	51.2	57.1	62.4	59.4	53.7	46.2	38.4	32.9		
		MCDB	39.8	38.1	48.3	55.7	65.7	69.2	75.5	75.8	70.1	57.0	45.4	38.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	22.2	22.2	24.6	24.9	23.7	28.4	29.2	27.6	25.4	22.1	20.6		
		MCDBR	22.2	27.4	29.4	35.7	34.2	31.5	35.6	37.8	38.4	33.6	27.3	22.7		
		MCWBR	15.3	18.8	16.9	17.8	14.5	13.7	14.7	14.3	15.8	17.2	16.2	15.4		
		MCDBR	21.9	26.4	28.0	33.1	30.8	27.8	30.0	32.9	35.4	32.2	26.6	22.5		
	5% WB	MCWBR	15.5	18.6	16.8	17.0	13.6	13.3	14.7	13.4	15.0	17.1	16.2	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.262	0.285	0.325	0.338	0.334	0.337	0.352	0.313	0.281	0.258	0.234		
	taud		2.427	2.473	2.464	2.366	2.384	2.440	2.447	2.388	2.475	2.532	2.489	2.438		
	Ebn at Noon		258	286	296	292	292	290	279	281	274	255	251			
	Edh at Noon		19	23	28	35	36	34	34	34	28	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		388	702	1115	1534	1829	2011	2194	1751	1240	767	421	294		
	RadStd		30	64	85	105	135	132	109	111	112	84	38	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page